Naive Bayes Classifier:

⇒ Supervised learning alg.
⇒ based on bayes theorem.
⇒ used for solving classification problems.
⇒ It is a 'probabilistic classifier': predicts based on the prob of an object.
⇒ 'Naive' because it assumes that the occurance of a certain feature is independent of the occurance of other features.

ex: if the fruit is identified on the bases of color, shape & taste; red, round, sweet is recognised as apple.
Hence, each feature individually contributes to identify that it is an apple w/o depending on eachother.

(posterior prob) likelihood prior prob

$$P(A|B) = \frac{P(B|A)\,P(A)}{P(B)}$$

$P(B)$ → evidence

Bayes Theorem

$P(A|B)$ ⇒ conditional prob of A given B
$P(B|A)$ ⇒ ~ of B ~ A
$P(A)$ ⇒ prob of event A
$P(B)$ ⇒ prob of event B.

Advantages:

* fast and efficient ⇒ req very lil training of data.
  ⇒ performs well with Large datasets.

* high-dimen data ⇒ can handle many features (attbs) without performance issues.

* handles missing data well ⇒ it uses probs, so it can still make predictions if some features are missing.

* simple, easy to implement ⇒ Basic prob theory: 'B. Theorem'.

Disadvantages:

* assumes all features are independent i.e rarely true in real world data.

* cannot be used for complex decision making: deep learning models (N.Nets)
  ensemble methods (R.forest)

* Zero-frequency problem: if a category is not p in the training data, the model assigns zero probability to that category.
  sol^n: laplace Smoothing can be used to avoid Z.F.P.

* Not suitable for continuous data.

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

NAIVE BAYES CLASSIFIER Example - 1

(i) calc tot prob

$P(PlayTennis = yes) = 9/14 = .64$

$P(PlayTennis = no) = 5/14 = .36$

(ii) make tables for attrs v/s pred

Outlook	Y	N
sunny	2/9	3/5
overcast	4/9	0
rain	3/9	2/5

Tempreature		
hot	2/9	2/5
mild	4/9	2/5
cool	3/9	1/5

Humidity	Y	N
high	3/9	4/5
normal	6/9	1/5

Windy		
Strong	3/9	3/5
Weak	6/9	2/5

NAIVE BAYES CLASSIFIER – Example -1

$\langle Outlook = sunny, Temperature = cool, Humidity = high, Wind = strong \rangle$

(iii) formula

$$\boxed{v_{NB} = \underset{v_j \in \{yes,no\}}{\operatorname{argmax}} P(v_j) \prod_i P(a_i|v_j)}$$

$$= \underset{v_j \in \{yes,no\}}{\operatorname{argmax}} P(v_j) \quad P(Outlook = sunny|v_j)P(Temperature = cool|v_j)$$

$$\cdot P(Humidity = high|v_j)P(Wind = strong|v_j)$$

$$v_{NB}(yes) = P(yes)\ P(sunny|yes)\ P(cool|yes)\ P(high|yes)\ P(strong|yes) = .0053$$

$$v_{NB}(no) = P(no)\ P(sunny|no)\ P(cool|no)\ P(high|no)\ P(strong|no) = .0206$$

(iv) validity

$$v_{NB}(yes) = \frac{v_{NB}(yes)}{v_{NB}(yes)+v_{NB}(no)} = 0.205 \qquad v_{NB}(no) = \frac{v_{NB}(no)}{v_{NB}(yes)+v_{NB}(no)} = 0.795$$

$+ = 1$

No	Color	Legs	Height	Smelly	Species
1	White	3	Short	Yes	M
2	Green	2	Tall	No	M
3	Green	3	Short	Yes	M
4	White	3	Short	Yes	M
5	Green	2	Short	No	H
6	White	2	Tall	No	H
7	White	2	Tall	No	H
8	White	2	Short	Yes	H

New Instance

(Color=Green, legs=2, Height=Tall, and Smelly=No)

NAIVE BAYES CLASSIFIER EXAMPLE - 2

(i) $P(M) = \frac{4}{8} = 0.5 \quad P(H) = \frac{4}{8} = 0.5$

(ii)

Color	M	H
White	2/4	3/4
Green	2/4	1/4

Legs	M	H
2	1/4	4/4
3	3/4	0/4

Height	M	H
Tall	3/4	2/4
Short	1/4	2/4

Smelly	M	H
Yes	3/4	1/4
No	1/4	3/4

NAIVE BAYES CLASSIFIER - EXAMPLE - 2

$P(M) = \frac{4}{8} = 0.5 \quad P(H) = \frac{4}{8} = 0.5$

Color	M	H
White	2/4	3/4
Green	2/4	1/4

Legs	M	H
2	1/4	4/4
3	3/4	0/4

Height	M	H
Tall	3/4	2/4
Short	1/4	2/4

Smelly	M	H
Yes	3/4	1/4
No	1/4	3/4

(iii)

$p(M|New\ Instance) = p(M) * p(Color = Green|M) * p(Legs = 2|M) * p(Height = tall|M) * p(Smelly = no\ |M)$

$p(M|New\ Instance) = 0.5 * \frac{2}{4} * \frac{1}{4} * \frac{3}{4} * \frac{1}{4} = 0.0117$

$p(H|New\ Instance) = p(H) * p(Color = Green|H) * p(Legs = 2|H) * p(Height = tall|H) * p(Smelly = no\ |H)$

$p(H|New\ Instance) = 0.5 * \frac{1}{4} * \frac{4}{4} * \frac{2}{4} * \frac{3}{4} = 0.047$

Example No.	Color	Type	Origin	Stolen?
1	Red	Sports	Domestic	Yes
2	Red	Sports	Domestic	No
3	Red	Sports	Domestic	Yes
4	Yellow	Sports	Domestic	No
5	Yellow	Sports	Imported	Yes
6	Yellow	SUV	Imported	No
7	Yellow	SUV	Imported	Yes
8	Yellow	SUV	Domestic	No
9	Red	SUV	Imported	No
10	Red	Sports	Imported	Yes

NAIVE BAYES CLASSIFIER EXAMPLE - 3

(i) $p(Yes) = \frac{5}{10} = 0.5$

$p(No) = \frac{5}{10} = 0.5$

Color	Yes	No
Red	3/5	2/5
Yellow	2/5	3/5

(ii)

Type	Yes	No
Sports	4/5	2/5
SUV	1/5	3/5

Origin	Yes	No
Domestic	2/5	3/5
Imported	3/5	2/5

(iii) $New\ Instance = (Red, SUV, Domestic)$

$P(Yes|New\ Instance) = p(Yes) * P(Color = Red|Yes) * P(Type = SUV|Yes) * P(Origin = Domestic|Yes)$

$P(Yes|New\ Instance) = \frac{5}{10} * \frac{3}{5} * \frac{1}{5} * \frac{2}{5} = \frac{3}{125} = 0.024$

$P(No|New\ Instance) = p(No) * P(Color = Red|No) * P(Type = SUV|No) * P(Origin = Domestic|No)$

$P(No|New\ Instance) = \frac{5}{10} * \frac{2}{5} * \frac{3}{5} * \frac{3}{5} = \frac{9}{125} = 0.072$

Naïve Bayes Classifier – Solved Example 5

Record	A	B	C	Class
1	0	0	0	+
2	0	0	1	−
3	0	1	1	−
4	0	1	1	−
5	0	0	1	+
6	1	0	1	+
7	1	0	1	−
8	1	0	1	−
9	1	1	1	+
10	1	0	1	+

A	0	1
+	2/5	3/5
-	3/5	2/5

B	0	1
+	4/5	1/5
-	3/5	2/5

C	0	1
+	1/5	4/5
-	0/5	5/5

$New \Rightarrow P(A = 0, B = 1, C = 0)$

$$P(+|New) = \frac{P(+) * P(A = 0|+) * P(B = 1|+) * P(C = 0|+)}{\cancel{P(A = 0, B = 1, C = 0)}}$$

$$= 0.5 * \frac{2}{5} * \frac{1}{5} * \frac{1}{5} = 0.008$$

+

$$P(-|New) = \frac{P(-) * P(A = 0|-) * P(B = 1|-) * P(C = 0|-)}{\cancel{P(A = 0, B = 1, C = 0)}}$$

$$= 0.5 * \frac{3}{5} * \frac{2}{5} * \frac{0}{5} = 0.0$$

$P(+|New) > P(-|New)$

Applns:

* medical data classification
* Spam filtering.
* Sentiment analysis.
* recommendation systems.